

Evaluating Frontier LLM Translation Capability for Lakota

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Abstract

We evaluate four frontier large language models on bidirectional Lakota-English translation using 200 sentence pairs from the New Lakota Dictionary. Two experimental conditions are compared: a baseline with near-deterministic decoding and no extended thinking, and a second condition with extended thinking enabled at maximum provider settings. The best model (Gemini 3.1 Pro) achieves a mean chrF++ of 59.4 on Lakota→English and 42.6 on English→Lakota. Extended thinking improves scores by 1-7 chrF++ points. No model produces reliable translation in either direction. An LLM semantic judge applied to the Lakota→English direction reveals that actual semantic equivalence ranges from 6% (GPT-5.2) to 60% (Gemini), diverging substantially from chrF++ scores. Diacritic normalization analysis shows models produce roughly correct base characters but place diacritical marks inconsistently. All results and evaluation code are publicly available at <https://github.com/robotson/lakota-translation-benchmark>.

1 Introduction

Lakota (*Lakǰótiyapi*) is a Siouan language spoken primarily on reservations in North and South Dakota. UNESCO classifies it as critically endangered (Moseley, 2010), with fewer than 2,000 first-language speakers, most over 65 years of age. Language revitalization efforts are ongoing across multiple tribal and academic institutions. The Lakota

Language Consortium (LLC) publishes the New Lakota Dictionary (NLD) and associated pedagogical materials; the NLD serves as the source for this study’s evaluation corpus.

Contemporary large language models (LLMs) support translation across dozens of languages, but the extent of training data for any given language is not disclosed. Users may reasonably expect similar quality across supported languages. For critically endangered languages like Lakota—where digitized text is scarce and no large-scale parallel corpus exists—this expectation is untestable without direct evaluation.

This paper reports the results of a direct evaluation. We tested four frontier LLMs on 200 Lakota-English sentence pairs in both translation directions, under two experimental conditions designed to test the impact of extended thinking (chain-of-thought reasoning) on translation quality. We report chrF++ and BLEU scores, diacritic normalization analysis, model self-reported confidence, and refusal behavior. Related work on LLM translation of low-resource languages includes the MTOB benchmark (Tanzer et al., 2024), which evaluated frontier LLMs on Kalamang translation using in-context learning from reference grammars, and the AmericasNLP shared tasks (Mager et al., 2021; Ebrahimi et al., 2023), which benchmark MT systems on indigenous American languages but do not include Lakota. The only prior Lakota-specific language model is LakotaBERT (Parankusham et al., 2025), a RoBERTa-based model

trained for masked language modeling. To our knowledge, no prior study has evaluated zero-shot frontier LLM translation quality for Lakota.

Lakota presents specific challenges for LLM translation. It is polysynthetic, with extensive verbal morphology including person marking, aspect, mood, and evidentiality encoded through affixes and enclitics. The Standard Lakota Orthography (SLO) uses Latin script with diacritical marks: caron-marked consonants (č, š, ž, ě, ě), ogoneks on vowels for nasalization (ą, ı, ı), acute accent for stress, and the glottal stop marker ('). The velar nasal is written with a separate character, ŋ. These diacritical marks distinguish phonemes—for example, *h* and *h̃* represent different consonants—making them significant for both translation accuracy and character-level evaluation metrics.

2 Methodology

2.1 Evaluation Corpus

The corpus consists of 200 sentence pairs sourced from published Lakota learning materials for research purposes. Pairs were drawn from conversational examples in the New Lakota Dictionary (NLD) (Ullrich, 2008), published by the Lakota Language Consortium (LLC), and selected from entries appearing under dictionary headwords. The first author filtered pairs for naturalness and conversational relevance. Each pair contains one Lakota sentence and one English translation. Examples range from simple greetings (*Tókheškhe yaúŋ he?* / “How are you?”) to multi-clause sentences with cultural context (*Wóčhičhiyakiŋ kte héčhiŋ.* / “I want to talk to you.”). The corpus is conversational register throughout.

The NLD uses the Standard Lakota Orthography (SLO), developed by the LLC. Multiple orthographic conventions have been in use across Lakota language communities, including the Buechel missionary system and the or-

thography used at Sinte Gleska University (White Hat, 1999). Standardization of Lakota orthography remains a contested issue within Lakota communities (Hauff, 2020). Our use of NLD materials was practical—it is the most accessible published collection of Lakota sentence pairs with English translations available to us—and should not be read as an endorsement of any particular orthographic standard or institutional approach to language documentation.

Each pair includes a conversational score (4–8) assigned during curation, where higher values indicate more everyday, conversational language and lower values indicate more formal or structurally complex sentences. The corpus also records the headword context from which each pair was drawn. The same 200 pairs were used for all models and both experimental conditions.

2.2 Models

Four frontier models from three providers were evaluated: Gemini 3.1 Pro Preview (Google), Claude Opus 4.6 (Anthropic), Claude Sonnet 4.6 (Anthropic), and GPT-5.2 (OpenAI). All models were accessed through litellm (v1.63), a unified API wrapper, to minimize implementation variance across providers. Selection criteria: frontier models from major providers, all supporting structured JSON output with schema enforcement.

2.3 Experimental Conditions

Two conditions were run on the same corpus:

Baseline: Temperature 0, no extended thinking. Maximum output tokens: 1,024. This condition uses near-deterministic decoding and represents baseline model capability.

Thinking: Extended thinking enabled at maximum provider settings. Maximum output tokens: 16,384 (to accommodate thinking token overhead). Claude models used

thinking.budget_tokens = 8,192; GPT-5.2 used reasoning_effort = high; Gemini used thinkingLevel = high.

A caveat applies to Gemini: thinking is enabled by default on Gemini 3.1 Pro and cannot be disabled. The baseline condition therefore already included default-level thinking for Gemini. The baseline→thinking comparison for Gemini isolates only the effect of increasing thinking from default to high, not of enabling thinking.

For Claude models, enabling extended thinking fixes decoding at temperature 1 (temperature cannot be set independently while thinking is enabled). For GPT-5.2, higher-reasoning configurations do not support sampling parameters such as temperature; although the thinking runs completed, the API client likely ignored the temperature=1 parameter when reasoning_effort was set to high. Accordingly, baseline→thinking comparisons should be interpreted as “baseline vs. maximum supported reasoning settings” rather than a clean temperature-controlled ablation.

2.4 Structured Output Schema

All models produced structured JSON output enforced via provider-native schema validation with three fields: translation (best translation or empty string), confidence (0.0–1.0), and refusal_reason (string or null).

The system prompt was identical across all models and conditions:

You are a translation system. Translate the input text. Respond as JSON with three fields: “translation” (your best translation, or empty string if you cannot translate), “confidence” (a number from 0.0 to 1.0 representing your confidence in the translation quality), “refusal_reason” (null if you attempted a translation, or a brief explanation if you cannot translate).

User prompts followed the format “Translate from Lakota to English: {text}” and “Translate from English to Lakota: {text}” for the two directions.

2.5 Metrics

chrF++ (character n-gram F-score) was the primary metric, computed via

SacreBLEU (Popović, 2015) with default parameters: character n-gram order 6, word n-gram order 2, $\beta = 2$. chrF++ is preferred over BLEU for morphologically rich languages because it operates at the character level and does not penalize valid morphological variants as harshly as word-level metrics. **BLEU** was computed as a secondary metric for comparability.

Diacritic-normalized chrF++ was computed for English→Lakota translations by stripping Unicode combining marks (categories Mn and Mc) from both candidate and reference via NFD decomposition before scoring. NFD decomposition ensures canonical equivalence between precomposed characters (e.g., U+01CE, ě) and combining sequences (e.g., a + U+030C) before mark removal. This normalization removes carons, ogoneks, and stress marks in SLO; the character ŋ (eng) and the glottal stop marker (ʔ) are unaffected. The delta between raw and normalized chrF++ isolates orthographic variation attributable to these diacritical marks from broader lexical and morphological differences.

LLM semantic judgment was used to evaluate Lakota→English translations, where chrF++ is known to penalize valid paraphrases against single references. We initially ran BERTScore (Zhang et al., 2020) (roberta-large) to capture semantic matches that chrF++ misses, but it was uninformative: F1 ranged from 0.906 to 0.959 across all models regardless of actual translation quality. At the pair level, GPT-5.2’s translation of “Do you have a car?” as “Do you have a horse?” received BERTScore 0.975. We instead used a separate LLM (Gemini 3 Flash Preview) to judge whether each hypothesis-reference pair conveys the same meaning, following the GEMBA framework (Kocmi and Federmann, 2023) but simplified to English-to-English semantic equivalence. Because both strings are English, the judge requires no knowledge of Lakota—it evaluates only

whether the model’s English output preserves the meaning of the English reference. Each pair was rated equivalent, partially_equivalent, or not_equivalent. All Lakota→English translation attempts across 8 model×condition combinations were judged.

Self-reported confidence (0.0–1.0) was extracted from the structured output.

2.6 Variance

A pilot study was conducted before the full thinking evaluation to assess run-to-run variance under the thinking condition’s decoding settings. Twenty sentence pairs were sampled and each model was run 3 times on both directions (480 API calls). Cross-run standard deviation ranged from 2.6 (Opus L→E) to 5.8 (GPT-5.2 E→L). Model rankings were stable across all three runs (Appendix A). Based on these results, the full evaluation was conducted as a single run per condition.

2.7 Procedure

Each evaluation run called the model API once per (pair, direction) combination: 200 pairs × 2 directions = 400 calls per model, 1,600 calls per condition. A 1.5-second delay was inserted between calls. API timeout was set to 60 seconds for baseline and 180 seconds for thinking. Failed calls were retried up to 3 times with exponential backoff. All evaluations were conducted in February–March 2026. Results were written to per-model JSONL files with per-pair metadata.

3 Results

3.1 Translation Quality

Table 1 reports the best-condition results for each model. For Gemini, baseline and thinking scores are nearly identical (§3.2); thinking is reported here for consistency since it includes confidence data. For all other models, thinking is reported as the higher-scoring condition.

Gemini leads in both directions. Opus and Sonnet score within 2 chrF++ points of each other on L→E (Sonnet slightly ahead), but Opus leads Sonnet by 7 points on E→L. GPT-5.2 trails in both directions. Every model scores substantially higher on Lakota→English than English→Lakota. The gap ranges from 6.4 points (GPT-5.2) to 19.1 points (Sonnet). This asymmetry is consistent across all models: Lakota→English (producing English from Lakota input) scores higher than English→Lakota (producing Lakota from English input) for every model.

Standard deviations are large—22–28 points for L→E and 13–22 points for E→L—indicating that per-pair quality varies widely. Some pairs score 100.0 (exact character match to reference) while others score below 10. Table 2 reports the distribution of scores across chrF++ thresholds and tail behavior (10th percentile).

Table 3 stratifies chrF++ by the corpus’s conversational score (§2.1). For Gemini L→E, the most conversational pairs (score 6) average 81.9 chrF++ while more complex pairs (score 4) average 52.7—a 29-point gap. The pattern holds across all models and both directions, suggesting that performance is concentrated on high-frequency phrasebook-like items.

Even the best model (Gemini L→E) produces translations scoring ≥80 chrF++—roughly usable quality—on only 24% of pairs. For E→L, only Gemini exceeds 5% at this threshold. If “reliable” is operationalized as ≥60 chrF++ on a majority of pairs, no model qualifies in either direction.

3.2 Effect of Extended Thinking

Table 4 compares baseline (near-deterministic decoding, no extended thinking) and thinking (maximum supported reasoning settings) for each model.

Thinking improved chrF++ across all model-direction pairs, with E→L gains

Table 1: Translation quality by model and direction (best condition). N < 200 reflects refusals, empty responses, or API errors excluded from scoring (§3.5). σ is across sentence pairs, not across runs.

Model	Dir	N	chrF++	σ (pairs)	Median	BLEU
Gemini 3.1 Pro	L→E	199	59.4	27.6	56.8	43.5
Claude Opus 4.6	L→E	200	45.9	25.0	43.5	27.2
Claude Sonnet 4.6	L→E	189	46.1	25.9	44.3	28.1
GPT-5.2	L→E	194	30.0	21.8	21.4	15.4
Gemini 3.1 Pro	E→L	199	42.6	22.1	38.9	24.7
Claude Opus 4.6	E→L	200	34.3	16.0	31.8	18.8
Claude Sonnet 4.6	E→L	196	27.0	13.2	24.2	15.7
GPT-5.2	E→L	184	23.6	12.5	20.5	14.3

Table 2: Score distribution by chrF++ threshold (thinking condition). Percentages over scored translations; P10 = 10th percentile.

Model	Dir	≥80	≥60	≥40	P10
Gemini	L→E	24%	47%	72%	21.6
Opus	L→E	12%	25%	56%	15.9
Sonnet	L→E	12%	25%	55%	16.0
GPT-5.2	L→E	5%	10%	23%	10.4
Gemini	E→L	7%	20%	47%	17.2
Opus	E→L	1%	6%	29%	16.0
Sonnet	E→L	1%	1%	15%	12.9
GPT-5.2	E→L	1%	1%	8%	12.1

Table 3: Mean chrF++ by conversational score (thinking condition). Score 4 = more formal/complex; score 6 = most conversational. Scores 7–8 excluded (n=6 combined).

Model	Dir	Sc. 4	Sc. 5	Sc. 6
Gemini	L→E	52.7	58.2	81.9
Opus	L→E	41.6	41.0	65.6
Sonnet	L→E	40.3	45.1	64.1
GPT-5.2	L→E	27.9	23.7	45.7
Gemini	E→L	37.1	42.7	61.3
Opus	E→L	31.2	32.7	45.6
Sonnet	E→L	24.7	26.2	35.2
GPT-5.2	E→L	22.2	21.5	30.4

(2.2–6.9 points, 5–29% relative) consistently larger than L→E gains (0.7–3.6 points, 2–14% relative). On L→E, three models showed only marginal improvement—Sonnet +0.7, Opus +0.9, Gemini +1.0—all within the variance measured in the pilot study. GPT-5.2 showed a larger L→E gain of +3.6. For Gemini, the small L→E delta is consistent with the fact that Gemini’s baseline already included default-level thinking.

Table 4: Baseline→thinking comparison

Model	Dir	Base	Think	Δ
Opus 4.6	L→E	45.0	45.9	+0.9
Opus 4.6	E→L	27.4	34.3	+6.9
Sonnet 4.6	L→E	45.4	46.1	+0.7
Sonnet 4.6	E→L	24.8	27.0	+2.2
GPT-5.2	L→E	26.4	30.0	+3.6
GPT-5.2	E→L	18.3	23.6	+5.3
Gemini	L→E	58.4	59.4	+1.0
Gemini	E→L	40.4	42.6	+2.2

Table 5: Diacritic normalization, English→Lakota

Model	Raw	Norm.	Δ
Gemini 3.1 Pro	42.6	48.7	+6.1
Claude Opus 4.6	34.3	40.8	+6.5
Claude Sonnet 4.6	27.0	32.3	+5.3
GPT-5.2	23.6	27.2	+3.6

Gemini’s E→L delta (+2.2) suggests that even with default thinking already active, the higher reasoning setting provides some benefit on the harder generation task.

The improvement is larger on E→L than L→E for all models, suggesting that reasoning overhead is more useful for generation (English→Lakota) than for the reverse direction.

3.3 Diacritic Normalization

Table 5 reports raw and diacritic-normalized chrF++ for English→Lakota (thinking condition).

Stripping diacritics adds 3.6–6.5 chrF++ points for E→L. The gap indicates that models produce roughly correct consonant and vowel sequences but place diacritical marks—

Table 6: Self-reported confidence (thinking condition)

Model	L→E	E→L
Gemini 3.1 Pro	0.93 ($\sigma=0.06$)	0.83 ($\sigma=0.11$)
Claude Opus 4.6	0.71 ($\sigma=0.17$)	0.46 ($\sigma=0.15$)
Claude Sonnet 4.6	0.70 ($\sigma=0.15$)	0.32 ($\sigma=0.11$)
GPT-5.2	0.55 ($\sigma=0.12$)	0.45 ($\sigma=0.11$)

stress, nasalization, and caron-marked consonants—inconsistently. Multiple orthographic conventions are in active use for Lakota (§2.1); models may have been trained on text in several of these systems, and the prompt did not specify an orthographic standard.

3.4 Confidence

Table 6 reports mean self-reported confidence from the thinking condition.

The confidence ranking largely matches the chrF++ ranking for L→E: Gemini > Opus $\approx$ Sonnet > GPT-5.2. All models report lower confidence for E→L than L→E, correctly reflecting the direction’s greater difficulty.

Gemini reports the highest confidence in both directions (0.93 L→E, 0.83 E→L) and also achieves the highest chrF++ scores, so its confidence is directionally accurate. However, 0.83 confidence on E→L where chrF++ is 42.6 represents significant overconfidence in absolute terms.

GPT-5.2’s E→L confidence (0.45) is higher than Sonnet’s (0.32) despite GPT-5.2 scoring lower on actual translation quality (23.6 vs 27.0 chrF++). The confidence ranking for E→L does not perfectly match the quality ranking.

3.5 Refusals and Failures

Table 7 reports non-translation outcomes from the thinking condition.

GPT-5.2 refused the most translations overall (22 total, 16 on E→L). Refusal reasons typically cited inability to produce reliable Lakota text. Sonnet produced 11 empty responses on L→E (the model returned an empty translation string with no refusal reason) and refused 2 E→L translations. Opus pro-

Table 7: Non-translation outcomes (thinking condition). Ref = refusal, Emp = empty response, Err = API error.

Model	L→E		E→L	
	Ref	Emp	Ref	Emp
GPT-5.2	6	0	16	0
Sonnet 4.6	0	11	2	2
Opus 4.6	0	0	0	0
Gemini	1	0	0	1

duced complete translations on all 200 pairs in both directions. Gemini had one L→E refusal and one E→L empty response.

Refusals are concentrated on E→L (16 of 22 GPT refusals, both Sonnet refusals), consistent with models being less willing to generate in a language they have limited training data for.

Non-translation outcomes shifted between conditions. At baseline, Sonnet refused 12 E→L translations and GPT-5.2 refused 4; with thinking enabled, GPT-5.2’s refusals increased to 22 while Sonnet’s refusals decreased to 2 but empty responses appeared (11 L→E, 2 E→L). Opus produced complete translations under both conditions. Refusal patterns depend on experimental setup—temperature, thinking overhead, or both—not only on model capability.

3.6 Semantic Equivalence

Table 8 reports LLM semantic judgments for Lakota→English translations under both conditions. The judge rated each hypothesis-reference pair as equivalent, partially_equivalent, or not_equivalent.

The judge reveals that chrF++ is a reasonable proxy for models with higher equivalence rates—Gemini’s chrF++ of 58.4 corresponds to 60.4% semantic equivalence—but misleading for models with lower rates. GPT-5.2’s chrF++ of 26.4 implies some partial overlap, but only 6% of its translations are semantically correct. The remaining 87% are fluent English that does not preserve the source meaning (Appendix B).

Table 8: LLM semantic judge results, L→E. Equiv = semantically equivalent to reference; Partial = partially equivalent; Not = not equivalent.

Model	Cond.	Equiv	Partial	Not
Gemini	base	60.4%	23.4%	16.2%
Gemini	think	61.3%	23.1%	15.6%
Opus 4.6	base	31.5%	18.0%	50.5%
Opus 4.6	think	37.0%	21.0%	42.0%
Sonnet 4.6	base	29.0%	16.0%	55.0%
Sonnet 4.6	think	30.7%	23.8%	45.5%
GPT-5.2	base	6.0%	7.0%	87.0%
GPT-5.2	think	13.4%	9.8%	76.8%

Across models, 16–24% of translations are *partially equivalent*—capturing the core meaning but missing elements or shifting emphasis. This reflects partial meaning preservation that binary scoring would obscure.

Extended thinking provides the largest benefit to the weakest model: GPT-5.2 improves from 6.0% to 13.4% equivalent ($2.2\times$), while Gemini barely changes (60.4% to 61.3%). This is consistent with the chrF++ thinking deltas (§3.2): reasoning overhead helps most when the model has partial knowledge to surface.

4 Discussion

The LLM judge results (§3.6) demonstrate that chrF++ alone cannot reliably characterize model capability on low-resource translation. For Gemini, chrF++ of 58.4 closely tracks 60.4% semantic equivalence—the paraphrase penalty is modest when the model produces semantically correct output. For Claude, chrF++ of ~ 45 is slightly generous compared to 30–37% equivalence. For GPT-5.2, chrF++ of 26.4 still implies some overlap, but only 6% of translations are semantically correct—the remaining 87% are fluent English that does not mean the right thing. Although BERTScore was uninformative as an absolute metric, the per-pair Pearson correlation between chrF++ and BERTScore descends in lockstep with actual semantic equivalence: Gemini $r=0.91$, Opus $r=0.86$, Sonnet $r=0.83$,

GPT $r=0.79$. When the two metrics agree at the pair level, the model is doing something coherent; when they disagree, the model is producing fluent nonsense. This metric-agreement signal is computable without an LLM judge and could serve as a lightweight hallucination check for other low-resource evaluations.

For context, frontier models achieve chrF++ scores of 65–70 on high-resource pairs such as English–German (Jiao et al., 2023); Gemini’s 59.4 on Lakota→English is deceptively close to this range, but the 42.6 on English→Lakota reveals the actual capability gap. The AmericasNLP shared tasks report chrF++ of 16–28 for other indigenous languages (Ebrahimi et al., 2023), making Gemini’s Lakota→English score notably higher than this range—though the comparison is not apples-to-apples, as those tasks evaluate fine-tuned NMT systems translating *into* the indigenous language. For English→Lakota, no model exceeds chrF++ 42.6. These scores are inadequate for unsupervised translation use.

Extended thinking provides a consistent but small improvement (1–7 chrF++ points). This represents a 2–29% relative improvement that does not change the practical assessment: models that produced unreliable translations without thinking produce slightly less unreliable translations with thinking. The improvement is larger on the harder E→L direction, suggesting reasoning overhead helps more on the lower-scoring direction. Gemini’s base-line→thinking delta of +1.0 on L→E (§3.2) is important context: since increasing Gemini’s thinking from default to high had minimal effect on Lakota→English scores, Gemini’s lead over the other models is primarily attributable to its base model weights or training data, not to reasoning overhead. The larger E→L delta (+2.2) suggests that higher reasoning settings provide some benefit on the harder gen-

eration task even when default thinking is already active.

The wide per-pair variance ($\sigma = 22\text{--}28$ chrF++ for L→E) likely reflects the distribution of Lakota text in training data. Common phrases such as *Máza-ska luhá he?* (“Do you have any money?”) score at or near 100.0 across all models, suggesting these phrases appear verbatim in training data. Culturally specific constructions involving kinship terms, complex verb morphology, or ceremonial language score near zero. The conversational score stratification (Table 3) confirms this pattern quantitatively: for every model and direction, score-6 pairs substantially outperform score-4 pairs. This suggests that model performance is concentrated on high-frequency phrasebook-like items rather than reflecting general capability across Lakota sentence types.

5 Conclusion

No frontier LLM tested in March 2026 can reliably translate Lakota. The best Lakota→English performance (chrF++ 59.4) corresponds to 60.4% semantic equivalence on conversational sentences, but the worst model (GPT-5.2) achieves only 6% equivalence despite producing fluent English throughout. The best English→Lakota performance (chrF++ 42.6) is inadequate for unsupervised use. Extended thinking provides a modest improvement of 1–7 chrF++ points but does not change the practical assessment. Diacritic normalization analysis shows models produce roughly correct base characters but place diacritical marks inconsistently, possibly reflecting orthographic heterogeneity in training data.

Ethics Statement

The evaluation corpus was drawn from the New Lakota Dictionary (NLD), a commercially published reference work. The corpus is used for research evaluation and is not redistributed. We

acknowledge that orthographic standardization is a contested issue within Lakota communities (Hauff, 2020), and that our use of NLD materials reflects availability rather than endorsement of any particular orthographic standard or institutional approach. Future work should evaluate against materials in other actively used orthographies, ideally in collaboration with language programs at tribal colleges and universities.

Limitations

Each pair has one reference translation. Lakota allows multiple valid translations of a given English sentence and vice versa. chrF++ against a single reference underestimates model quality for valid translations that differ from the reference; the LLM judge partially mitigates this by accepting valid paraphrases, but cannot fully solve it. All semantic judgments come from a single judge model (Gemini 3 Flash Preview); inter-judge agreement is not assessed. The corpus consists of constructed conversational examples from a dictionary; performance on narrative text or spontaneous speech may differ. The baseline→thinking comparison reflects a bundled change from near-deterministic decoding to maximum supported reasoning settings rather than a clean ablation.

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Author Contributions

L.R. designed the study, curated the evaluation corpus, ran all evaluations, and made all methodological decisions.

Data Availability

Code and aggregate results are available at <https://github.com/robotson/>

[lakota-translation-benchmark](#). The evaluation corpus was sourced from the New Lakota Dictionary and is not redistributed; see `data/example_pairs.json` for the data schema. Aggregate results are provided in `results/comparison.csv` and `results/llm_judge_summary.csv`. All code requires only API keys and `pip install litellm sacrebleu python-dotenv bert-score`.

A Pilot Variance

Table 9: Pilot variance (20 pairs, 3 runs each)

Model	Dir	chrF++	σ
Gemini 3.1 Pro	L→E	63.1	3.3
Gemini 3.1 Pro	E→L	33.6	3.4
Claude Opus 4.6	L→E	45.1	2.6
Claude Opus 4.6	E→L	29.8	4.8
Claude Sonnet 4.6	L→E	42.7	4.8
Claude Sonnet 4.6	E→L	20.8	4.6
GPT-5.2	L→E	25.5	5.7
GPT-5.2	E→L	15.9	5.8

B Fluent Hallucination Examples

Table 10 illustrates GPT-5.2’s pattern of producing fluent, structurally plausible English that does not preserve the source meaning. Several of these score well on chrF++ because the surface overlap is high—the errors are surgical: a reversed subject, a single wrong noun, a shifted pronoun.

Table 10: GPT-5.2 L→E translations judged not_equivalent despite high chrF++.

Reference	GPT-5.2 output	chrF++
I love my mother and father.	My mother and father love me.	71.0
Do you have a car?	Do you have a horse?	66.4
Do you like Indian food?	Do you like Lakota food?	56.0
Do you like pastry?	Do you like the story?	57.2
How many horses do you have?	How many horses do we have?	42.0

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